

Numerical Analysis of SRP and Lunisolar Third-Body Perturbations on GEO Satellite Orbital Elements using Gauss Variational Equations

Brandon Lopez¹ and Ryan Omori²

Abstract—GEO satellites are subject to perturbation forces that cause their orbital elements to deviate from the ideal Keplerian motion over time. Unlike low Earth orbit satellites, where atmospheric drag and Earth’s oblateness dominate, GEO satellites are primarily influenced by solar radiation pressure (SRP) and lunisolar third-body gravitational effects. This work investigates the influence of these perturbations using a body-centric Earth-centered formulation and the Gauss Variational Equations (GVE). Four simulation cases are considered: a pure Keplerian baseline, SRP only, lunisolar gravity only, and the full combined perturbation model. Solar radiation pressure is modeled using the cannonball approximation, while lunar and solar gravitational effects are represented through third-body perturbation accelerations. Numerical propagation is performed in both Cartesian and orbital-element space to evaluate the evolution of the osculating orbital elements and validate the perturbation formulation. The results are compared to quantify secular and periodic variations in the orbital elements and to assess the relative importance of each perturbation source. The study is expected to show that lunisolar gravity dominates the long-term evolution of GEO orbits, while SRP contributes measurable periodic variations, providing insight into the perturbation environment relevant to GEO station-keeping and mission design.

I. INTRODUCTION

Geostationary Earth orbit (GEO) satellites provide critical services for communications, weather monitoring, navigation, and national security applications. Operating at an altitude of approximately 35,786 km above Earth’s equator, GEO spacecraft maintain an orbital period equal to Earth’s rotational period, allowing them to remain fixed relative to a point on the Earth’s surface. Because many GEO satellites remain operational for decades, accurate prediction of their long-term orbital behavior is essential for station-keeping, mission planning, and collision avoidance.

In the ideal two-body problem, a satellite follows a Keplerian orbit governed solely by Earth’s gravitational attraction, and its orbital elements remain constant in time. Real spacecraft, however, are subjected to a variety of perturbation forces that cause deviations from this idealized motion. While low Earth orbit satellites are primarily affected by atmospheric drag and Earth’s oblateness, these effects become significantly weaker at GEO altitude. Instead, solar radiation pressure (SRP) and the gravitational attraction of external bodies, particularly the Moon and Sun, become dominant sources of long-term orbital evolution [1], [2].

Solar radiation pressure results from the momentum transfer associated with solar photons impacting the spacecraft surface. Although the resulting acceleration is several orders of magnitude smaller than Earth’s gravitational acceleration, its cumulative influence over long durations can produce measurable changes in orbital elements. Studies comparing SRP effects on both low Earth orbit and geostationary satellites have shown that orbital parameters such as semi-major axis and eccentricity are particularly sensitive to solar radiation pressure, while inclination and right ascension of the ascending node experience smaller variations [3]. Likewise, the Moon and Sun continuously exert third-body gravitational forces that generate both periodic and secular variations in GEO trajectories. These perturbations are among the primary drivers of long-term station-keeping requirements and therefore must be accurately modeled when predicting spacecraft motion.

The accurate representation of SRP has been an active area of research in astrodynamics. The simplest approach is the cannonball model, which approximates the spacecraft as a spherical object characterized by an effective area-to-mass ratio and reflectivity coefficient. The resulting SRP acceleration used is

$$a_{srp} = \frac{P_{sun}}{c} \left(\frac{a_{sun}}{r_{sun}} \right)^2 \frac{C_r A_{srp}}{m} \hat{s}, \quad (1)$$

where P_{sun} is the solar flux at 1 AU, c is the speed of light, C_r is the reflectivity coefficient, A_{srp} is the effective cross-sectional area, m is the spacecraft mass, and $\hat{s}$ is the Sun-spacecraft unit vector. For GEO applications, the ratio a_{sun}/r_{sun} is approximately one since the spacecraft’s distance from the Sun differs negligibly from 1 AU. Therefore, the SRP acceleration may be approximated as

$$a_{srp} \approx \frac{P_{sun}}{c} \frac{C_r A_{srp}}{m} \hat{s}. \quad (2)$$

More advanced models account for spacecraft geometry, attitude, and surface optical properties. Ziebart [4] developed a generalized analytical SRP modeling framework capable of representing complex spacecraft geometries and secondary reflections, while Montenbruck et al. [5] demonstrated that improved box-wing SRP models significantly reduce orbit prediction errors for Galileo navigation satellites. More recently, Li et al. [1] showed that SRP is the third-largest perturbation acting on many cislunar spacecraft and represents a major source of orbit determination error in Earth-Moon missions, highlighting the growing importance of accurate SRP modeling for future space exploration.

¹B.Lopez is with Faculty of Mechanical & Aerospace Engineering, Mechatronics and Controls, University of California San Diego

²R. Omori is with Faculty of Mechanical & Aerospace Engineering, Mechatronics and Controls, University of California San Diego

To analyze these effects, it is advantageous to reformulate the equations of motion in terms of orbital elements rather than Cartesian position and velocity coordinates. The Gauss Variational Equations (GVE) provide an exact relationship between disturbing accelerations and the resulting rates of change of the classical Keplerian orbital elements. By directly relating perturbing forces to variations in the semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of periapsis, and true anomaly, the GVE provides a physically intuitive framework for studying orbital evolution under non-Keplerian forces [2].

The objective of this study is to investigate the influence of solar radiation pressure and lunisolar third-body gravitational perturbations on the long-term evolution of GEO satellite orbital elements. A body-centric Earth-centered formulation is used in which Earth provides the dominant gravitational attraction while SRP and third-body forces are treated as disturbing accelerations. Four simulation cases are considered: a pure Keplerian baseline, SRP-only perturbations, lunisolar gravity-only perturbations, and a combined perturbation model. Numerical propagation is performed using the Gauss Variational Equations, and the resulting orbital element histories are compared to quantify the relative contribution of each perturbation source. Particular attention is given to the evolution of eccentricity, inclination, and angular orbital elements, which are directly related to GEO station-keeping requirements and long-term mission performance.

II. PROBLEM STATEMENT

The objective of this study is to quantify the influence of solar radiation pressure (SRP) and lunisolar third-body gravitational perturbations on the long-term evolution of geostationary Earth orbit (GEO) satellites. The spacecraft is modeled as a massless particle moving in an Earth-centered inertial reference frame. Earth provides the dominant gravitational attraction, while the Moon, Sun, and solar radiation pressure act as perturbing accelerations.

The governing equations of motion may be written as

$$\ddot{\mathbf{r}} = -\frac{\mu_E}{r^3} \mathbf{r} + \mathbf{a}_{SRP} + \mathbf{a}_{Moon} + \mathbf{a}_{Sun}, \quad (3)$$

where $\mathbf{r}$ is the geocentric spacecraft position vector, μ_E is Earth's gravitational parameter, and $\mathbf{a}_{SRP}$, $\mathbf{a}_{Moon}$, and $\mathbf{a}_{Sun}$ represent the disturbing accelerations due to solar radiation pressure and the lunar and solar third-body gravitational effects, respectively [6].

The third-body perturbing acceleration generated by an external body of gravitational parameter μ_p is represented using the standard body-centric formulation

$$\mathbf{a}_d = -\mu_p \left(\frac{\mathbf{r} - \mathbf{r}_p}{|\mathbf{r} - \mathbf{r}_p|^3} + \frac{\mathbf{r}_p}{|r_p|^3} \right), \quad (4)$$

where $\mathbf{r}_p$ denotes the position vector of the perturbing body relative to Earth. Equation (4) is applied separately for the Moon and the Sun [7].

Rather than directly analyzing the Cartesian state variables, the equations of motion are transformed into the Gauss Variational Equations (GVE), which describe the evolution of

the classical orbital elements under disturbing accelerations. This formulation allows the influence of each perturbation source on the semi-major axis a , eccentricity e , inclination i , right ascension of the ascending node Ω , argument of periapsis ω , and true anomaly f to be examined directly.

To isolate the contribution of each perturbation source, four numerical experiments are considered:

- 1) Pure Keplerian motion
- 2) Solar radiation pressure only,
- 3) Lunisolar third-body gravity only,
- 4) Combined SRP and lunisolar perturbations.

The resulting orbital element histories are compared to determine the relative importance of each perturbation mechanism and to identify which orbital elements are most sensitive to long-term perturbation effects in GEO.

III. SOLUTION TECHNIQUES AND METHODOLOGY

A. Gauss Variational Equations

Rather than propagating the spacecraft state directly in Cartesian coordinates, the orbital motion is represented using the classical Keplerian orbital elements

$$[a, e, i, \Omega, \omega, f]^T \quad (5)$$

The evolution of these elements is governed by the Gauss Variational Equations (GVE), which relate disturbing accelerations resolved in the radial-transverse-normal (RTN) frame to the corresponding orbital element rates. The perturbing acceleration is decomposed as:

$$\mathbf{a}_d = R\hat{\mathbf{r}} + T\hat{\mathbf{t}} + N\hat{\mathbf{n}}, \quad (6)$$

where (R), (T), and (N) denote the radial, transverse, and normal acceleration components, respectively. These components are substituted into the GVE to determine the time evolution of the orbital elements throughout the numerical integration.

B. Lunisolar Third-Body Perturbations

The gravitational attraction of the Moon and Sun is modeled using the standard body-centric third-body formulation. The perturbing acceleration generated by an external body of gravitational parameter (μ_p) is:

$$\mathbf{a}_d = -\mu_p \left(\frac{\mathbf{r} - \mathbf{r}_p}{|\mathbf{r} - \mathbf{r}_p|^3} + \frac{\mathbf{r}_p}{|r_p|^3} \right), \quad (7)$$

Equation (7) is evaluated independently for the Moon and Sun, and the resulting accelerations are projected onto the RTN frame for use within the GVE.

C. Solar Radiation Pressure Model

Solar radiation pressure is modeled using the cannonball approximation, in which the spacecraft is represented by an equivalent sphere characterized by an effective area-to-mass ratio and reflectivity coefficient. The resulting acceleration is

$$\mathbf{a}_{SRP} = \frac{P_{sun}}{c} \left(\frac{a_{sun}}{r_{sun}} \right)^2 \frac{C_r A_{srp}}{m} \hat{\mathbf{s}}, \quad (8)$$

D. Orbit-Averaged SRP Formulation

To efficiently model the long-term effects of solar radiation pressure, the orbit-averaged SRP formulation developed by Guffanti et al. and based on the classical work of Cook is used. In this approach, the SRP acceleration is decomposed into averaged radial, transverse, and normal components denoted by (R_p) , (S_p) , and (W) , respectively.

The averaged orbital element rates due to SRP are given by

$$\dot{e}_{SRP} = \frac{3\eta}{2na} S_p, \quad (9)$$

$$\dot{i}_{SRP} = -\frac{3eW \cos \omega}{2na\eta}, \quad (10)$$

$$\dot{\Omega}_{SRP} = -\frac{3eW \sin \omega}{2na\eta \sin i}, \quad (11)$$

$$\dot{\omega}_{SRP} = -\frac{3\eta}{2nae} R_p - \dot{\Omega}_{SRP} \cos i, \quad (12)$$

$$\dot{M}_{SRP} = \frac{9eR_p}{2na} - \eta (\dot{\omega}_{SRP} + \dot{\Omega}_{SRP} \cos i), \quad (13)$$

where $\eta = \sqrt{1 - e^2}$ is the eccentricity factor and $n = \sqrt{\mu_E/a^3}$ is the mean motion. The true anomaly rate is obtained from the mean anomaly through the standard orbital element transformation. This averaged formulation captures the secular and long-periodic effects of SRP while avoiding the computational expense associated with modeling short-period oscillations directly.

E. Numerical Integration

The orbital element equations are numerically calculated using MATLAB's `ode113` variable-step integrator with relative and absolute tolerances of 10^{-10} . Four simulation cases are considered:

- 1) Pure Keplerian motion,
- 2) Solar radiation pressure only,
- 3) Lunisolar third-body gravity only,
- 4) Combined SRP and lunisolar perturbations.

Each case is propagated for one year beginning from a near-circular geostationary orbit. The resulting orbital element histories are compared to isolate the contribution of each perturbation source and to identify the orbital elements most sensitive to long-term GEO perturbations.

F. Model Validation

The numerical implementation was validated using the unperturbed Keplerian case. In the absence of solar radiation pressure and third-body perturbations, the Gauss Variational Equations reduce to the classical two body problem, for which all orbital elements remain constant in time except for the true anomaly. Therefore, the semi-major axis, eccentricity, inclination, right ascension of the ascending node, and argument of periapsis were expected to remain unchanged throughout the propagation which can be seen below in Table I.

TABLE I
KEPLERIAN VALIDATION RESULTS

Element	Initial Value	Final Value	Δ
a [km]	42164.000000	42164.000000	0.000000
e	0.001000	0.001000	0.000000
i [deg]	0.100000	0.100000	0.000000
Ω [deg]	0.000000	0.000000	0.000000
ω [deg]	0.000000	0.000000	0.000000

IV. RESULTS

The GEO satellite was propagated for one year using four dynamical models: an unperturbed Keplerian baseline, solar radiation pressure (SRP) only, lunisolar third-body gravity only, and the combined perturbation model.

A. Orbital Evolution in the GEO Environment

Figure 1 illustrates the propagated spacecraft trajectory for the simulation period of 1 year.

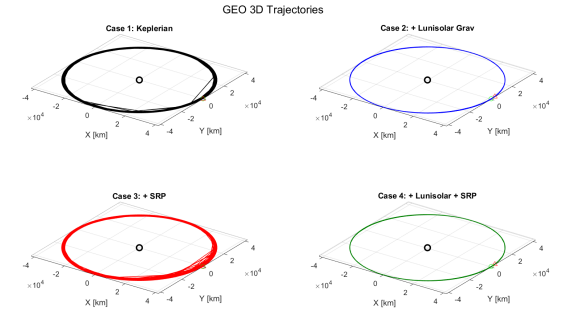


Fig. 1. Propagated GEO trajectory under the modeled perturbation environment.

B. Semi-Major Axis Evolution

Table II shows the variation in semi-major axis for all simulation cases over the 1-year propagation. The semi-major axis remains constant under pure Keplerian and SRP-only dynamics, confirming that neither perturbation drives secular growth in a . Lunisolar third-body gravity introduces periodic oscillations of approximately 2 km peak-to-trough at roughly monthly frequency, which dominate the combined case as well. SRP produces no measurable contribution to semi-major axis variation at GEO.

TABLE II
SEMI-MAJOR AXIS VARIATION AFTER ONE YEAR

Case	a_0 [km]	a_f [km]	Δa [km]
Keplerian	42164.000	42164.000	0.000
Lunisolar Gravity	42164.000	42162.036	-1.964
Solar Radiation Pressure	42164.000	42164.000	0.000
Lunisolar Gravity + SRP	42164.000	42162.036	-1.964

C. Eccentricity Evolution

Figure 2 presents the eccentricity evolution for all perturbation cases. SRP drives a slow secular growth reaching approximately 5×10^{-4} over one year, consistent with the

orbit-averaged formulation. Lunisolar gravity alone produces smaller oscillations centered near zero. The combined case exhibits both the secular trend from SRP and superimposed periodic modulation from lunisolar effects, resulting in the largest peak eccentricity of the three perturbed cases. This persistent increase in eccentricity is particularly important for GEO communication satellites because growing eccentricity causes the orbit to deviate from its nominal circular configuration, increasing station-keeping requirements and potentially degrading long-term mission performance if left uncorrected.

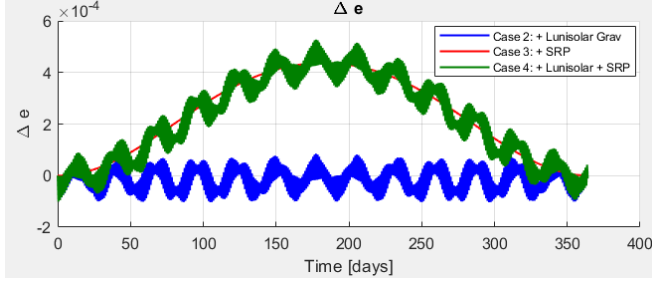


Fig. 2. Eccentricity evolution under the different perturbation models.

D. Inclination Evolution

Figure 3 shows the inclination variation throughout the simulation period. Inclination remains essentially unperturbed by SRP, as expected from the orbit-averaged equations where i_{SRP} vanishes for near-zero eccentricity. Lunisolar third-body gravity drives a monotonic secular increase of approximately 0.33° over one year, consistent with the well-known inclination drift that dominates GEO north-south station-keeping budgets. The combined case tracks the lunisolar-only result closely.

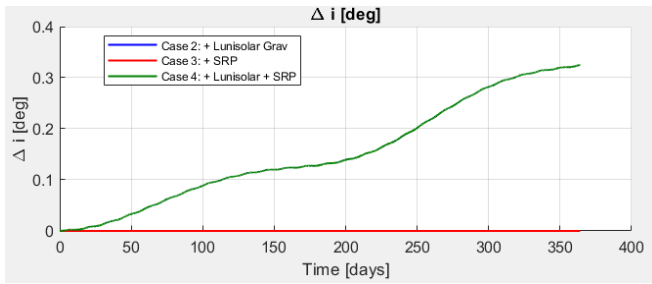


Fig. 3. Inclination evolution for each of the perturbation models.

E. Angular Orbital Element Evolution

1) *Right Ascension of the Ascending Node*: Figure 4 shows the variation in right ascension of the ascending node (RAAN). The RAAN exhibits a large secular drift driven entirely by lunisolar gravity, reaching approximately 75° over the one-year simulation. SRP contributes negligibly. This secular RAAN drift is a direct consequence of the out-of-plane lunisolar acceleration and represents one of the primary long-term perturbation signatures at GEO altitude.

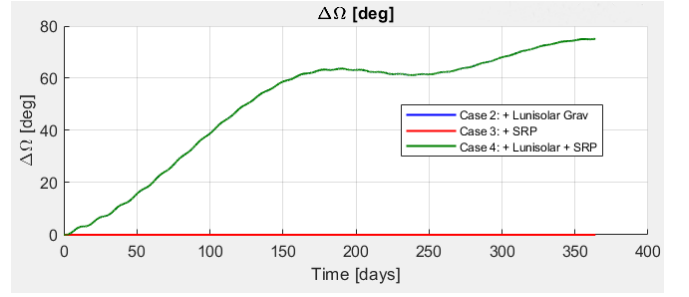


Fig. 4. Evolution of the right ascension of the ascending node.

2) *Argument of Periapsis*: Figure 5 presents the evolution of the argument of periapsis. The argument of periapsis shows distinct behavior between perturbation sources. SRP produces a slow positive drift peaking near 15° before returning toward zero, reflecting the orbit-averaged $\dot{\omega}_{\text{SRP}}$ dependence on eccentricity and the SRP geometry. Lunisolar gravity drives a large negative secular drift of nearly 80° over one year, which dominates the combined case and superimposes periodic oscillations on the secular trend.

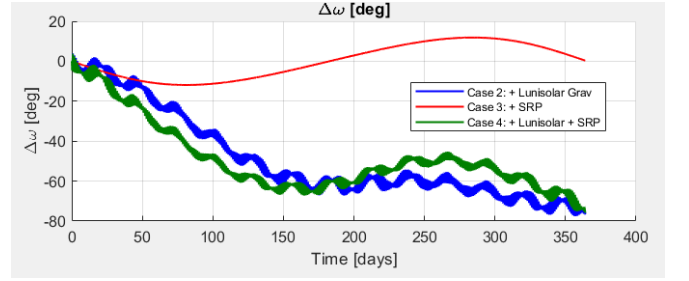


Fig. 5. Evolution of the argument of periapsis under the different perturbation models.

3) *True Anomaly*: Figure 6 shows the evolution of the true anomaly throughout the propagation interval. True anomaly deviations remain bounded within approximately $\pm 15^\circ$ for all perturbed cases, indicating that the perturbations do not accumulate significant along-track drift over one year at GEO. The periodic oscillations in the combined case arise from the interaction of SRP-driven eccentricity growth with the lunisolar perturbation of the orbit geometry.

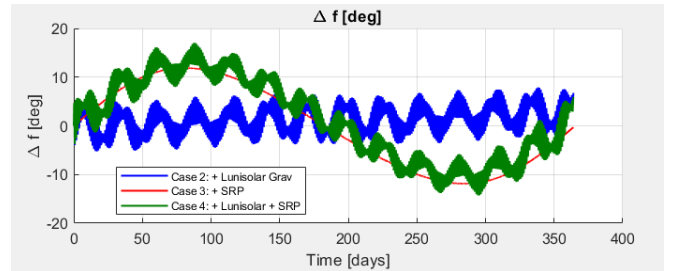


Fig. 6. True anomaly evolution for each perturbation model.

V. DISCUSSION

A. Interpretation of Simulation Results

The four simulation cases reveal a clear hierarchy in perturbation significance for GEO orbital evolution. Lunisolar third-body gravity dominates the long-term dynamics, driving secular growth in inclination of approximately 0.33° per year and a large secular RAAN drift exceeding 70° over the simulation interval, as shown in Figures 9 and 10. These results are consistent with the well-known GEO inclination drift that constitutes the primary driver of north-south station-keeping budget requirements in operational satellites. The argument of periaapsis also exhibits a large negative secular drift under lunisolar forcing, reflecting the out-of-plane gravitational torque exerted by the Moon and Sun on the near-equatorial GEO orbit plane.

Solar radiation pressure, by contrast, produces its most significant effects on eccentricity, as illustrated in Figure 8, consistent with the orbit-averaged analytical formulation of Cook adopted in this study [4]. The secular eccentricity growth driven by SRP reaches approximately 5×10^{-4} over one year. SRP produces negligible contributions to inclination and RAAN, as expected from the averaged equations where $\dot{i}_{SRP}$ and $\dot{\Omega}_{SRP}$ are proportional to eccentricity and therefore vanish for the near-circular initial conditions considered here. The combined perturbation case confirms that these two sources act largely independently, with lunisolar gravity governing out-of-plane evolution and SRP governing in-plane eccentricity growth.

Regarding the semi-major axis, all perturbed cases show no secular drift, as seen in Figure 7, which is a direct consequence of the orbit-averaged SRP formulation and the body-centric third-body model adopted. The analytical models of Cook, as presented by Guffanti et al. [4], explicitly yield $\langle \dot{a} \rangle = 0$ for both SRP and lunisolar perturbations in the averaged sense. While higher-order short-period oscillations in a are known to exist in real satellite data, these are filtered out by the orbit-averaging procedure and lie outside the scope of this secular perturbation analysis.

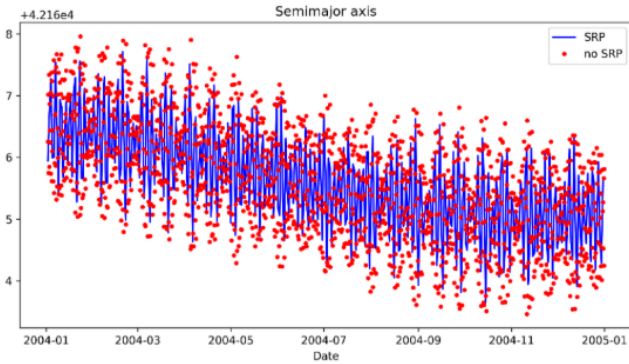


Fig. 7. Semi-major axis evolution from Pratiwi et al. [5] for the Telkom-1 GEO satellite. The shared downward trend in both the SRP and no-SRP cases suggests a common perturbation source not isolated in their figure, consistent with the result of the present orbit averaged model shown in Table II

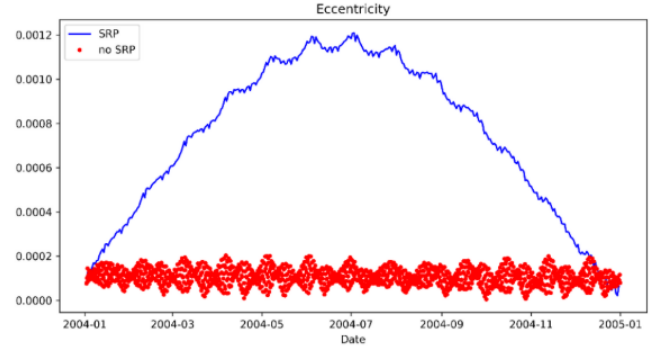


Fig. 8. Eccentricity evolution from Pratiwi et al. [5] for the Telkom-1 GEO satellite, showing SRP (blue) and no-SRP (red) cases over 2004. The secular growth and annual periodicity are qualitatively consistent with the simulation results of Figure 8.

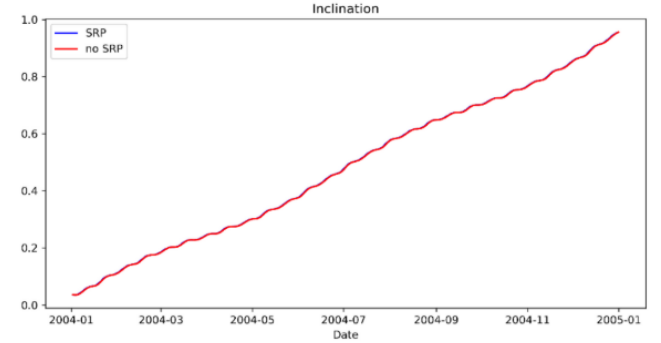


Fig. 9. Inclination evolution demonstrating the secular drift driven by lunisolar third-body gravity over the one-year simulation.

B. Comparison with Observed GEO Satellite Data

To contextualize these simulation results, a qualitative comparison is drawn with the observational study of Pratiwi et al. [5], which analyzed TLE-derived orbital element histories for the Telkom-1 GEO satellite over a one-year period using a spherical SRP model in Freeflyer. Their results provide an independent, observation-based reference against which the physical trends predicted by this simulation can be assessed.

For eccentricity, the agreement in character is strong. As shown in Figure 8, the simulation reproduces the key

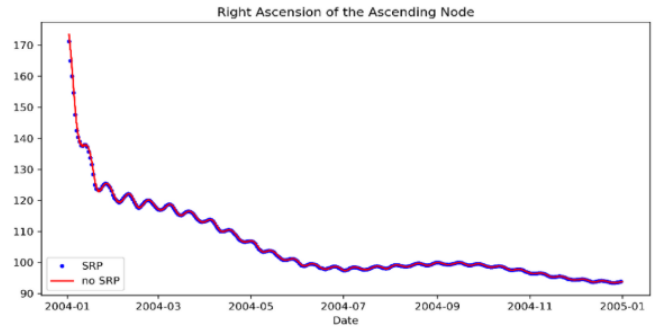


Fig. 10. Right ascension of the ascending node evolution showing the large secular drift attributable entirely to lunisolar perturbations.

qualitative behavior reported by Pratiwi et al.: eccentricity is significantly more sensitive to SRP than inclination or RAAN, with the SRP case producing a slow secular growth modulated by the evolving Sun–spacecraft geometry over the year. Both studies find that inclination and RAAN are insensitive to SRP, as confirmed by Figures 9 and 10, further supporting the physical consistency of the cannonball model across both studies.

The semi-major axis comparison warrants careful interpretation, and is illustrated alongside the Pratiwi et al. reference data in Figure 7. In Pratiwi et al., both the SRP and no-SRP cases exhibit a similar downward trend in a over the year, with SRP adding only short-period oscillations on top of this shared drift. The presence of this trend in both cases strongly suggests it is driven by a perturbation source common to both scenarios — most likely lunisolar third-body gravity, residual atmospheric drag, or station-keeping maneuvers embedded in the TLE data, none of which are isolated in their figure. This interpretation is consistent with our simulation result in Table II, in which the orbit-averaged lunisolar and SRP models both correctly yield $\langle \dot{a} \rangle = 0$ in the secular sense. The short-period oscillations visible in the Pratiwi et al. semi-major axis data represent sub-orbital-period variations that are intentionally averaged out in the Cook formulation underlying our propagator, and their absence in our results is expected rather than a modeling deficiency.

Collectively, these comparisons support the physical validity of the perturbation model developed in this study, while highlighting that real satellite data incorporates additional perturbation sources and operational effects — including atmospheric drag, geopotential harmonics, and active station-keeping — that are beyond the scope of this analytical framework.

VI. CONCLUSIONS

This study investigated the influence of SRP and lunisolar third-body gravitational perturbations on the long-term evolution of geostationary Earth orbit (GEO) satellites using the Gauss Variational Equations. Four simulation cases were considered to isolate the effects of SRP, lunisolar gravity, and their combined influence. The results showed that lunisolar gravity dominates the long-term evolution of inclination, right ascension of the ascending node, and argument of periapsis, while SRP primarily affects the eccentricity of the orbit.

Although the acceleration produced by solar radiation pressure is small compared to Earth’s gravitational attraction, its cumulative effect over long time periods produces measurable changes in eccentricity. This result is particularly important for communication satellites operating in GEO, where maintaining a nearly circular orbit is essential for reliable coverage, accurate pointing, and efficient station-keeping operations. The simulations demonstrated that SRP-driven eccentricity growth can become comparable to or larger than the variations produced by lunisolar gravity alone, highlighting the importance of accurately modeling non-gravitational perturbations in long-duration missions.

Comparison with published GEO satellite observations further emphasized the sensitivity of orbital predictions to the chosen SRP model. While the simplified cannonball approximation successfully reproduced the qualitative behavior of eccentricity evolution, previous studies have shown that spacecraft geometry, attitude, and surface optical properties can significantly influence the magnitude and direction of the SRP force. Consequently, inaccurate assumptions regarding spacecraft physical characteristics may lead to substantial discrepancies between theoretical predictions and experimentally observed orbital trajectories.

Overall, the results demonstrate that SRP should not be treated as a negligible perturbation for GEO applications. Accurate representation of solar radiation pressure is necessary for long-term orbit prediction, station-keeping analysis, and mission planning. Future work could incorporate higher-fidelity SRP models, spacecraft attitude dynamics, Earth shadowing effects, and geopotential perturbations to improve agreement with operational satellite observations and provide a more complete representation of the GEO perturbation environment.

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